

Ordinary Differential Equations

Differential Equations as Mathematical Models & Linear Models

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ENGR 213**

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Today's Lecture

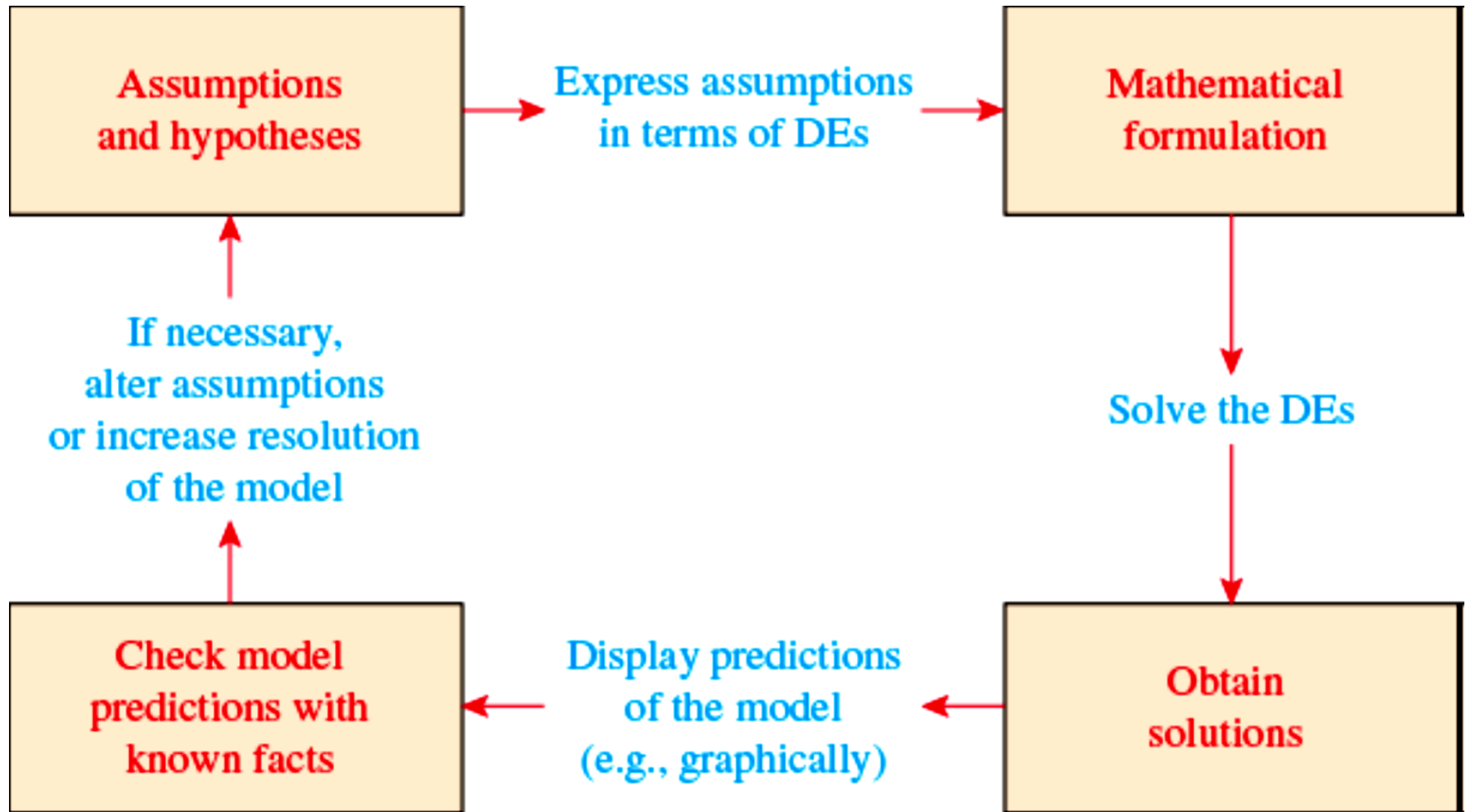
- Linear models
 - ✓ Growth/decay
 - ✓ Heating/cooling
 - ✓ Mixtures
 - ✓ Circuits

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 2.7**

- **Mathematical model:** a mathematical description of a real-world system or phenomenon, built for a specific purpose.
- **Model development:** identify variables that are responsible for changing the system, choose variables and the desired level of detail, and make reasonable assumptions or use relevant empirical laws.
- Because many assumptions involve rates of change, the resulting model often takes the form of a DE or system of DEs.
- Solve the mathematical model and compare its predictions with experimental data or known system behavior.
- If predictions are poor, refine the assumptions or increase the model resolution—usually at the cost of greater mathematical complexity.
- In time-dependent models, the solution describes the state and evolution of the system in the past, present, and future.

Steps in the modeling process



□ Growth/Decay

- Malthusian model of population growth: the rate at which population grows at a certain time is proportional to the total population of the at that time. In mathematical terms, if $P(t)$ denotes the total population at time t :

$$\frac{dP}{dt} \propto P \quad \longrightarrow \quad \frac{dP}{dt} = kP \quad (*\text{where } k \text{ is a constant of proportionality})$$

- Radioactive decay: the rate at which the nuclei of a substance decay is proportional to the amount of the substance (A) remaining at time t :

$$\frac{dA}{dt} \propto A \quad \longrightarrow \quad \frac{dA}{dt} = kA \quad (*\text{where } k \text{ is a constant of proportionality})$$

* For growth, $k > 0$, and in the case of decay, $k < 0$.

EXAMPLE 1 (Bacterial Growth)

A culture initially has P_0 number of bacteria. At $t = 1$ h the number of bacteria is measured to be $\frac{3}{2} P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.



□ Newton's Law of Cooling/Warming

- Newton's empirical law of cooling (or warming): the rate at which the temperature of a body changes (dT/dt) is proportional to the difference between the temperature of the body ($T(t)$) and the temperature of the surrounding medium (T_m):

$$\frac{dT}{dt} \propto (T - T_m) \longrightarrow \frac{dT}{dt} = k(T - T_m) \quad (k \text{ is a constant of proportionality})$$

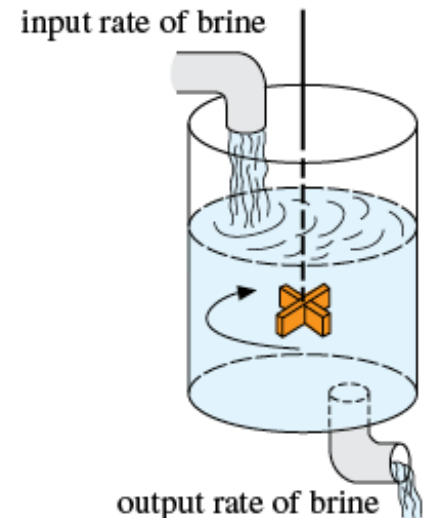
EXAMPLE 3 (Cooling of a Cake)

When a cake is removed from an oven, its temperature is measured at 300°F . Three minutes later its temperature is 200°F . How long will it take for the cake to cool off to a room temperature of 70°F ?



❑ Mixtures

- The mixing of two fluids sometimes gives rise to a linear first-order differential equation.
- For example, consider mixing of two brine solutions of different salt concentrations which gives rise to a first-order DE for the amount of salt contained in the mixture:



$$\frac{dA}{dt} = (\text{input rate of salt}) - (\text{output rate of salt})$$

* $A(t)$ denotes the amount of salt in the tank at time t

$$\frac{dA}{dt} = Ri_n - R_{out} \longrightarrow \frac{dA}{dt} = c_{salt,in}Q_{in} - c_{salt,out}Q_{out}$$

* Q denotes flowrate
* C_{salt} is concentration

Different scenarios: $Q_{in} = Q_{out}$ $Q_{in} < Q_{out}$ $Q_{in} > Q_{out}$

EXAMPLE 3 (Mixture of Two Salt Solutions)

A large tank holds 300 gallons of brine solution. A brine solution is being pumped into the tank at the rate of 3 gal/min, mixed with the solution there, and then the mixture is pumped out at the rate of 3 gal/min. The concentration of the salt in the inflow is 2 lb/gal. If there is 50 lb of salt dissolved initially in the 300 gallons, how much salt will be in the tank after a long time?



□ Series Circuits

- For a single-loop LRC-series circuit containing an inductor, resistor, and capacitor, Kirchhoff's second law states: *The impressed voltage $E(t)$ on a closed loop must equal the sum of the voltage drops in the loop.*

$i(t)$: current

$q(t)$: charge on a capacitor

L : inductance (henrys(h))

R : resistance (ohms (Ω))

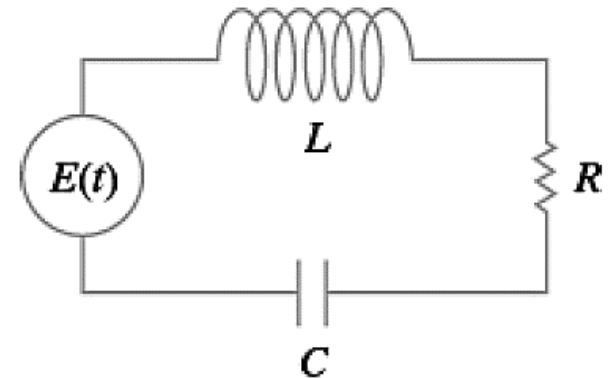
C : capacitance (farads(f))

Voltage drop

Inductor: $L \frac{di}{dt} = L \frac{d^2 q}{dt^2}$

Resistor: $iR = R \frac{dq}{dt}$

Capacitor: $\frac{1}{C} q$



LRC-series circuit

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$$

EXAMPLE 4 (LR-Series Circuit)

A 12-volt battery is connected to an LR-series circuit in which the inductance is 0.5 henry and the resistance is 10 ohms. Determine the current i if the initial current is zero.